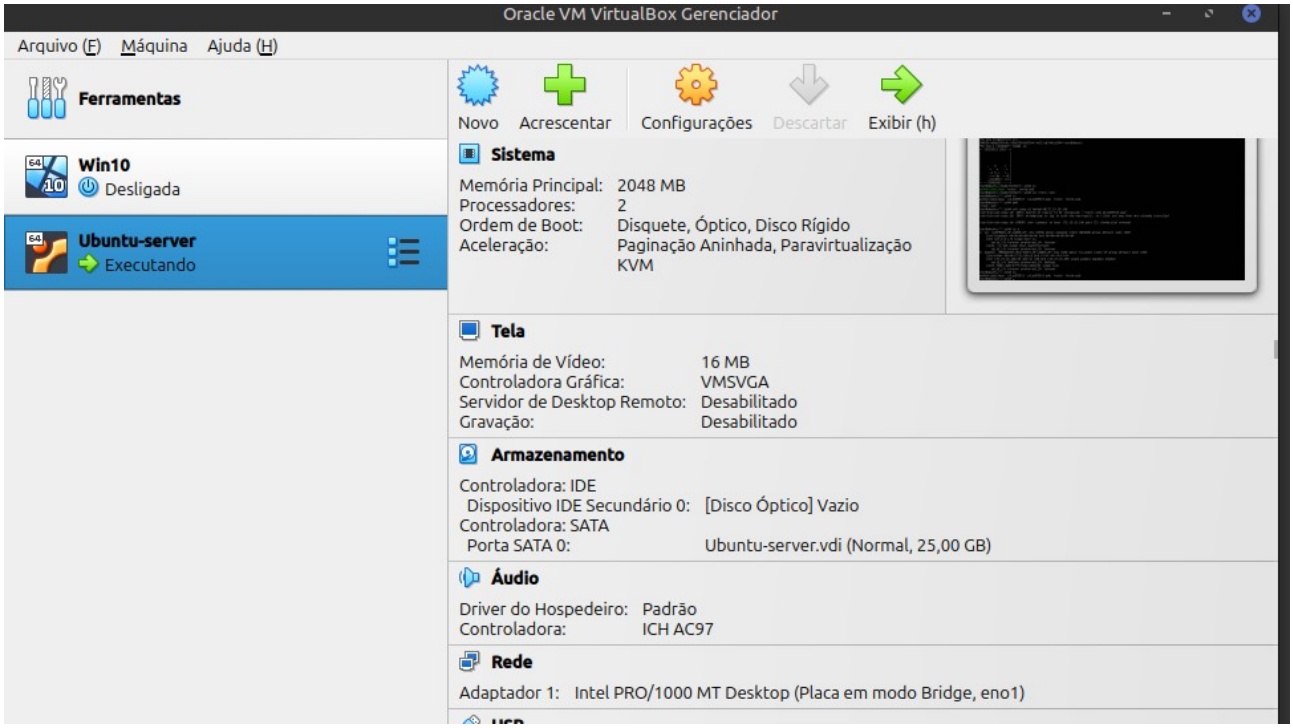
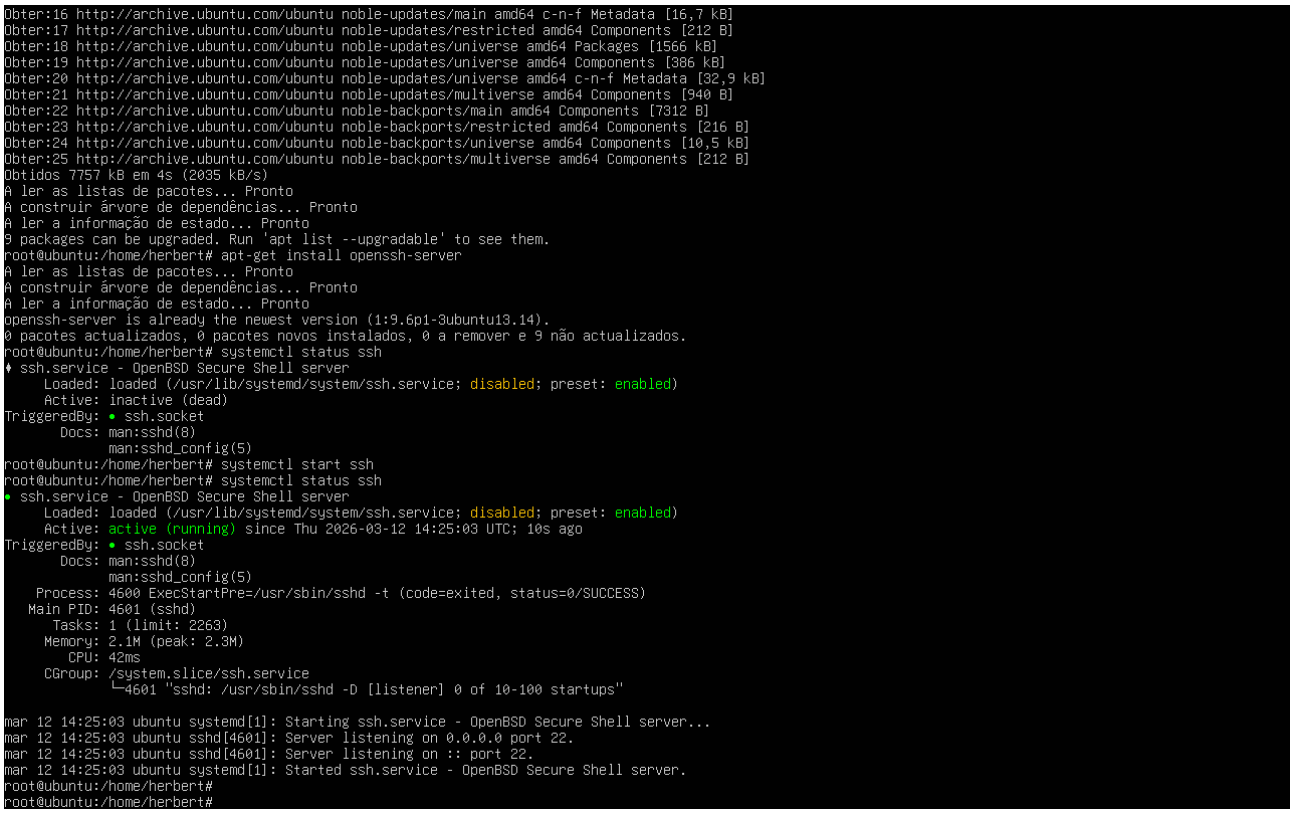
Atividade 01 – Conexão SSH

1 – Criação da máquina virtual:



2 – Instalação do servidor SSH

sudo apt install openssh-server



```
Obter:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16,7 kB]
Obter:17 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Obter:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1566 kB]
Obter:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Obter:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32,9 kB]
Obter:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Obter:22 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7312 B]
Obter:23 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Obter:24 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10,5 kB]
Obter:25 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Obtidos 7757 kB em 4s (2035 kB/s)
A ler as listas de pacotes... Pronto
A construir árvore de dependências... Pronto
A ler a informação de estado... Pronto
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ubuntu:/home/herbert# apt-get install openssh-server
A ler as listas de pacotes... Pronto
A construir árvore de dependências... Pronto
A ler a informação de estado... Pronto
openssh-server is already the newest version (1:9.6p1-3ubuntu13.14).
0 pacotes actualizados, 0 pacotes novos instalados, 0 a remover e 9 não actualizados.
root@ubuntu:/home/herbert# systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: inactive (dead)
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
root@ubuntu:/home/herbert# systemctl start ssh
root@ubuntu:/home/herbert# systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: active (running) since Thu 2026-03-12 14:25:03 UTC; 10s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 4600 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 4601 (sshd)
      Tasks: 1 (limit: 2263)
     Memory: 2.1M (peak: 2.3M)
        CPU: 42ms
     CGroup: /system.slice/ssh.service
             └─4601 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

mar 12 14:25:03 ubuntu systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
mar 12 14:25:03 ubuntu sshd[4601]: Server listening on 0.0.0.0 port 22.
mar 12 14:25:03 ubuntu sshd[4601]: Server listening on :: port 22.
mar 12 14:25:03 ubuntu systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
root@ubuntu:/home/herbert#
root@ubuntu:/home/herbert#
```

3 – Identificação do IP
ip a

```
root@ubuntu:/home/herbert# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2c:20:c8 brd ff:ff:ff:ff:ff:ff
    inet 172.19.18.106/24 metric 100 brd 172.19.18.255 scope global dynamic enp0s3
       valid_lft 4311sec preferred_lft 4311sec
    inet6 fe80::a00:27ff:fe2c:20c8/64 scope link
       valid_lft forever preferred_lft forever
root@ubuntu:/home/herbert# hostname -I
172.19.18.106
root@ubuntu:/home/herbert# _
```

4 – Criação das chaves SSH
ssh-keygen

```
root@ubuntu:/home/herbert# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2c:20:c8 brd ff:ff:ff:ff:ff:ff
    inet 172.19.18.106/24 metric 100 brd 172.19.18.255 scope global dynamic enp0s3
       valid_lft 4311sec preferred_lft 4311sec
    inet6 fe80::a00:27ff:fe2c:20c8/64 scope link
       valid_lft forever preferred_lft forever
root@ubuntu:/home/herbert# hostname -I
172.19.18.106
root@ubuntu:/home/herbert# ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:35oHqt8ZPrHHoMQRTIoLrWg6IVZ+WAwGygD/TGFDvOk root@ubuntu
The key's randomart image is:
+--[ED25519 256]--+
|+..==  o.        |
|+..o++ .o        |
|..o.+=.  .       |
| .o*=.  .        |
|+o.++. .S.       |
|=.  E   o.+.     |
|o      . oo*.    |
| .      ooo=+    |
|      .o. *+     |
+----[SHA256]-----+
root@ubuntu:/home/herbert# _
```

5 – Copia da chave pública
ssh-copy-id

```
root@ubuntu:/home/herbert/.ssh# ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519): /home/herbert/.ssh/teste
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/herbert/.ssh/teste
Your public key has been saved in /home/herbert/.ssh/teste.pub
The key fingerprint is:
SHA256:e86pS3OYrbi/1BoC2264zX28+K/+nZ1/xg7xMLq19H4 root@ubuntu
The key's randomart image is:
+--[ED25519 256]--+
|                 |
|                 |
|                 |
|                 |
|    .   S     +  |
|     +   *   . = |
|    ..o O.= . +..|
|    .+.* @o. = *E|
|    .o*oXBO=+ +=X|
+----[SHA256]-----+
root@ubuntu:/home/herbert/.ssh# ls
authorized_keys  teste  teste.pub
root@ubuntu:/home/herbert/.ssh# cd /root/.ssh/
root@ubuntu:~/.ssh# ls
authorized_keys  id_ed25519  id_ed25519.pub  teste  teste.pub
root@ubuntu:~/.ssh# pwd
/root/.ssh
root@ubuntu:~/.ssh# _
```

6 – Teste da conexão

```
herbert@ubuntu: ~
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
herbert@172.19.18.106's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'herbert@172.19.18.106'"
and check to make sure that only the key(s) you wanted were added.

root@DSCOTIGD23141:/home/herbert# ls
'Área de trabalho'
 Documentos
 Downloads
 Imagens
 libgl1-mesa-glx_23.0.4-0ubuntu1~22.04.1_amd64.deb
 libssl1.1_1.1.1f-1ubuntu2.23_amd64.deb
 lynis.log
 lynis-report.dat
 mintupgrade-2025-09-08T161634.log
 mintupgrade-2025-09-08T161719.log
 Modelos
 Músicas
 nano.18089.save
 pt
 Público
 Vídeos
'VirtualBox VMs'
root@DSCOTIGD23141:/home/herbert# herbert@172.19.18.106
herbert@172.19.18.106: comando não encontrado
root@DSCOTIGD23141:/home/herbert# ssh herbert@172.19.18.106
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of qui 12 mar 2026 18:51:42 UTC

  System load:  0.0               Processes:               119
  Usage of /:   40.8% of 11.21GB  Users logged in:         1
  Memory usage: 12%               IPv4 address for enp0s3: 172.19.18.106
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Manutenção de Segurança Expandida para Applications não está ativa.

7 as atualizações podem ser aplicadas imediatamente.
1 dessas atualizações é uma atualização de segurança padrão.
Para ver as actualizações adicionais corre o comando: apt list --upgradable

Ativar ESM Apps para poder receber possiveis futuras atualizações de segurança.
Consulte https://ubuntu.com/esm ou execute: sudo pro status


Last login: Thu Mar 12 18:42:37 2026 from 172.19.18.146
herbert@ubuntu:~$
quinta, março 12, 15:52
```